

Types of Agents

- ③ Goal-based Agent
- current state
 - context given : desirable destination
- Goal → Destination
- ↳ Searching
 - ↳ Planning

- ④ Utility-based Agent

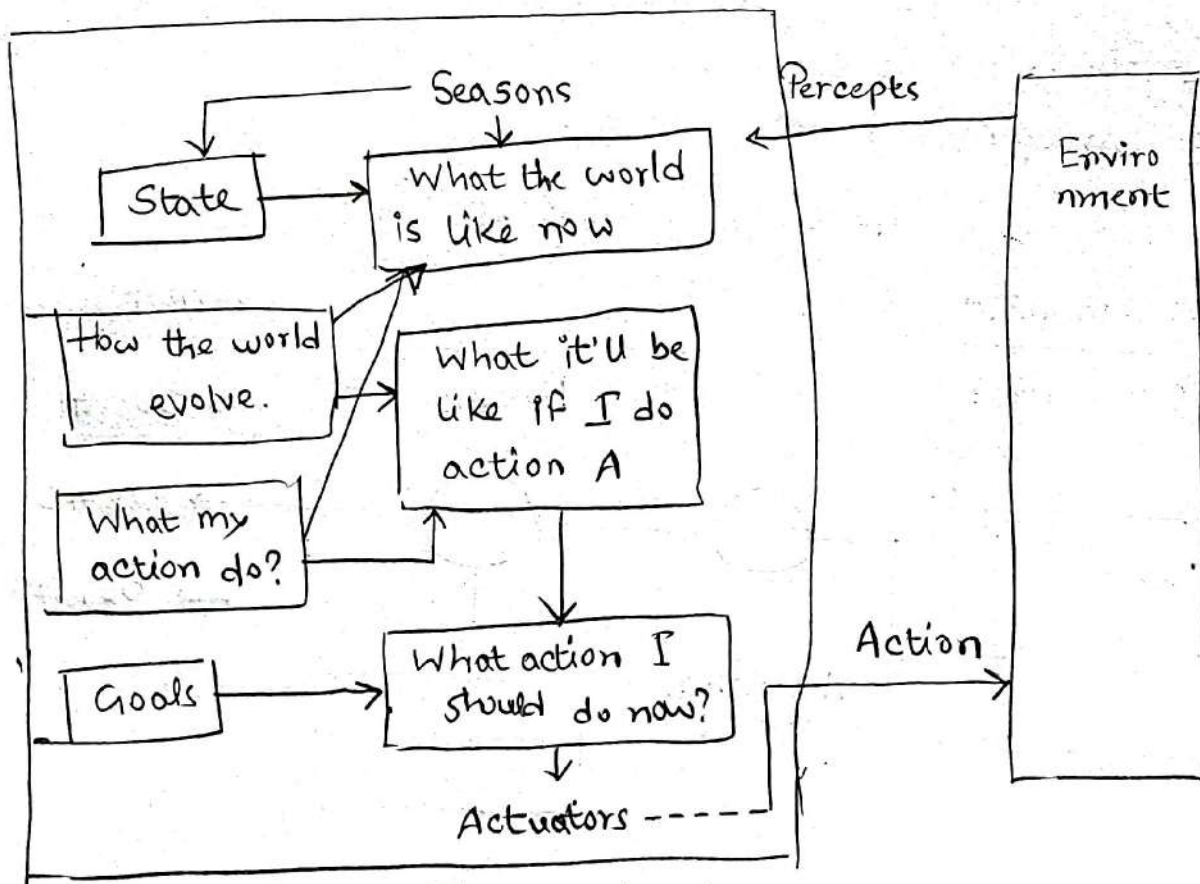
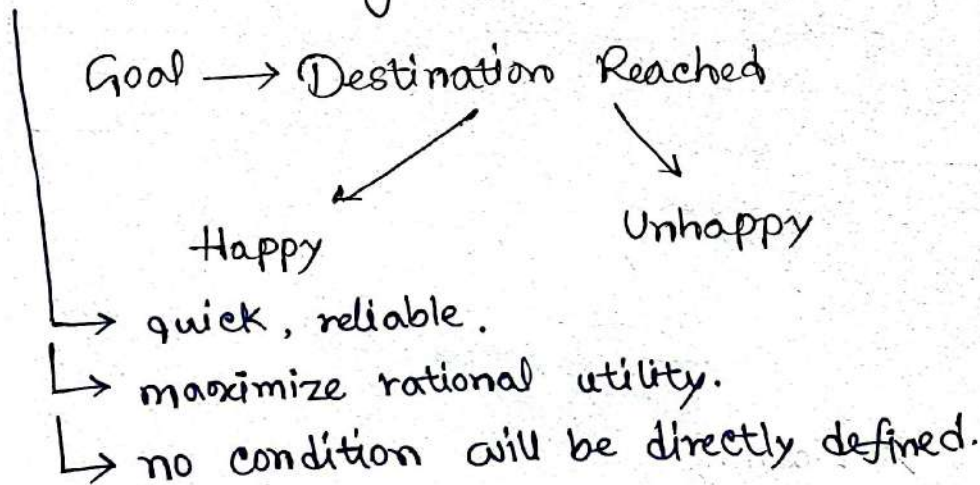
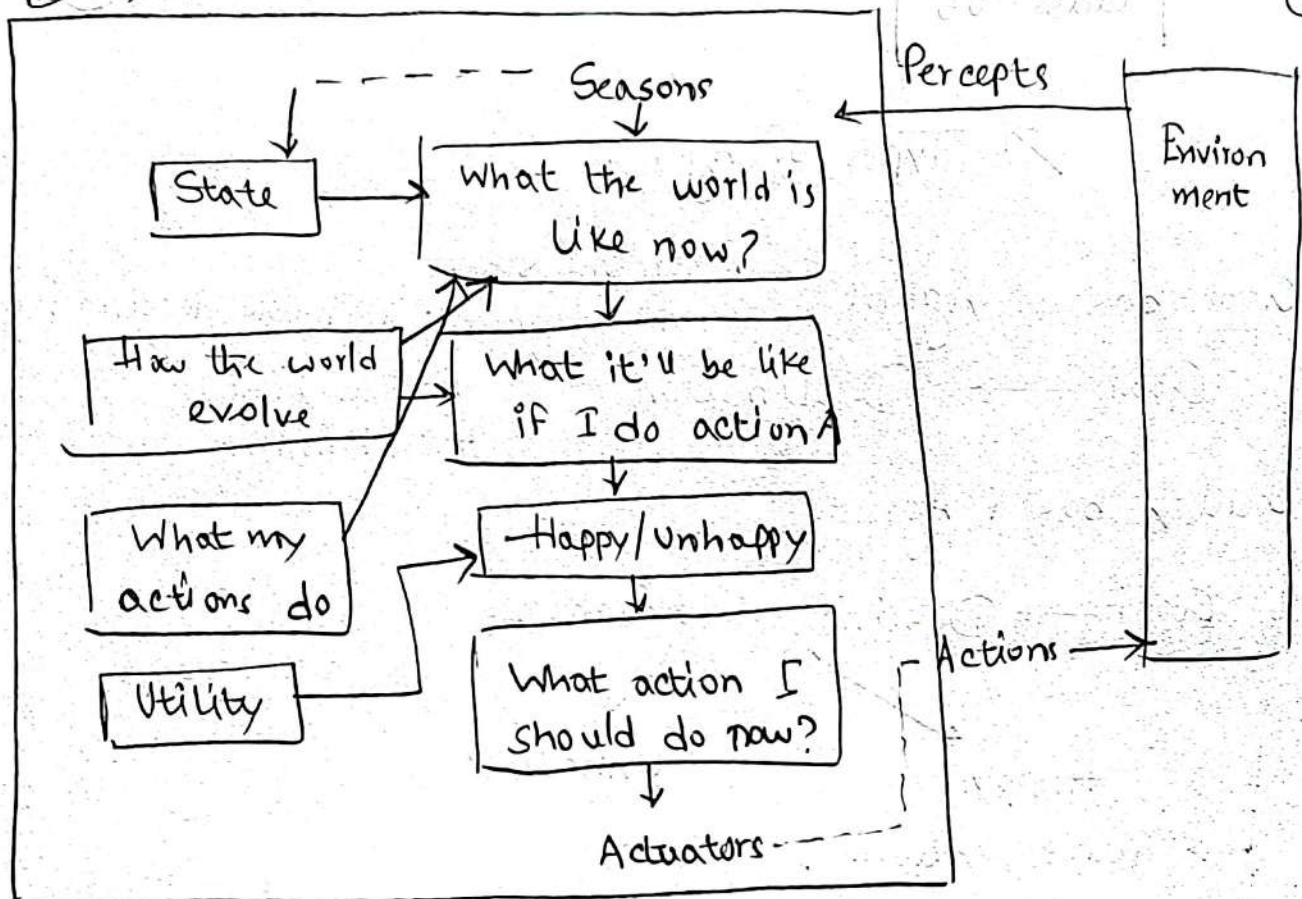


figure : Goal-based Agent

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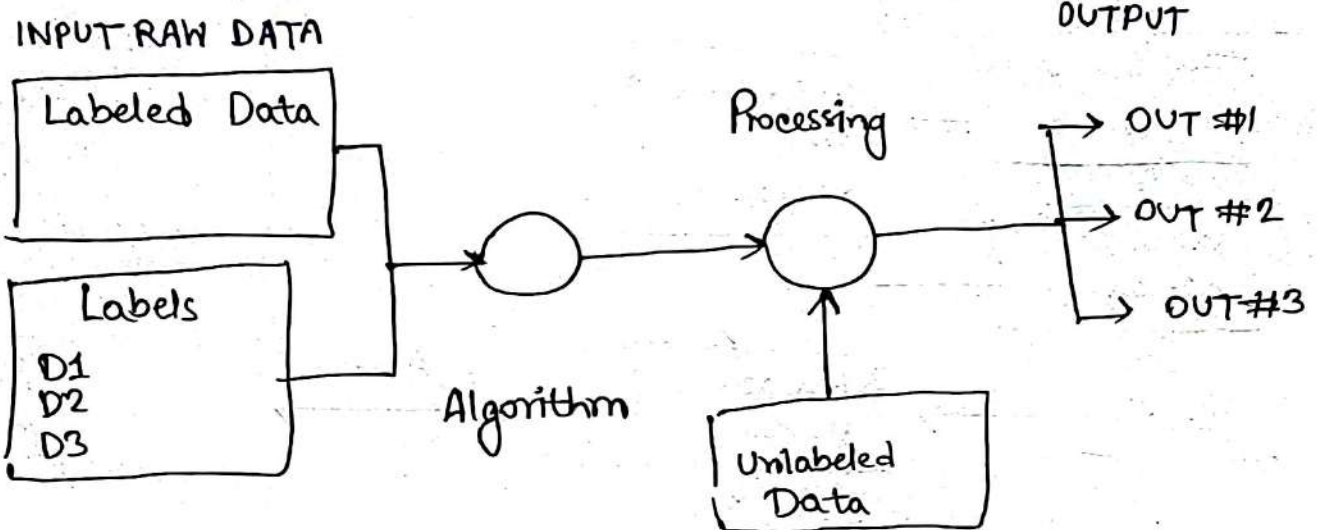
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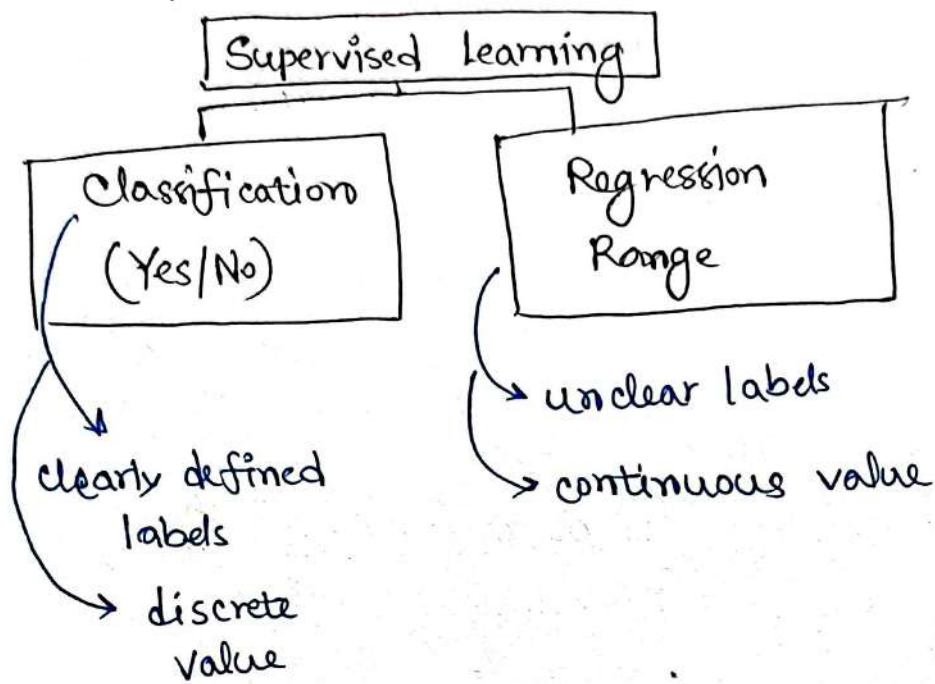
Learning

- ☒ Supervised
- ☒ Unsupervised

Supervised Machine Learning



Shayan | ③

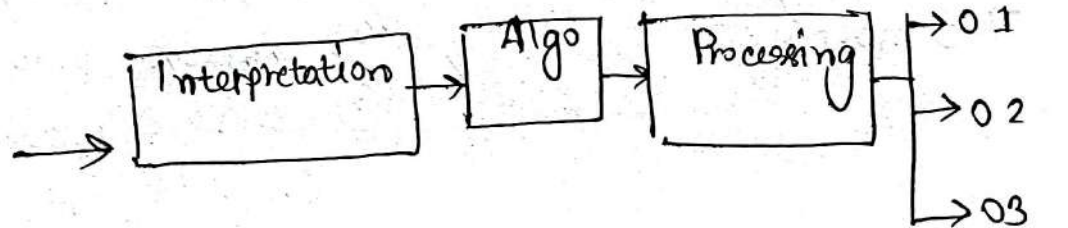


Linear Regression
Logistic Regression

Unsupervised Learning

INPUT RAW DATA

(Unlabeled Images)



- Unknown output
- No training dataset

- Clustering
- Dimensionality / Association
- (Supervised vs Unsupervised Learning) Comparison

Shayan | ④

ML Algorithms

Unsupervised

Clustering & Dimensionality
Reduction

CT ①

Topic: Lecture 1, 2

Monday (15.6.26)

Wednesday (17.6.26)

{ 7.5 marks }

Problem Solving & Planning

Four Things

① Define the problem precisely

② Analyze problem.

③ Isolate & represent the task

Knowledge that is necessary to
solve the problem. → search

④ Choose the best problem-

solving technique(s) &

apply it/them to the
particular problem.

Goal Formation

Problem
Formation

Solution

↗ Goal Formulation

Formulate Goal:

To be in Bucharest

Formulate Problem:

States: Various cities

Actions: Drive between cities

Find Solution:

By searching through states to find a goal.

- Find appropriate sequence of stops.

Multistate Problem → Not fully observable

Single state problem → fully observable

↗ Problem Formulation

5 Components

① Initial State: That the agent starts in.

② Actions: Description of possible actions available to the agent.

③ Transition Model: Description of what each action does specified by function, $result(s, a)$ that return the state that results from doing action 'a' in State 's'.

→ Together: the initial state, actions, transition model, implicitly define the State space of the problem,

Shayan | ⑥

iv) Goal Test: Determines whether the given state is a goal state.

v) Path Cost: A path in the state space is a 'seq' of states connected by sequence of actions.

↗ State Space Definition

Next class : Problem Solving